

Domenico De Giorgio

📍 Milan, Italy | 📞 +39 389-176-6712 | ✉️ domenico.degiorgio@icloud.com
🌐 Portfolio | 🐙 DomDegi | 🌐 Domenico De Giorgio

EDUCATION

Politecnico di Milano	M.Sc. High-Performance Computing Engineering <i>Parallel Computing, Advanced Computer Architectures, Numerical Linear Algebra, Partial Differential Equations</i>	Sep 2025 – Present
Politecnico di Milano	B.Sc. Engineering of Computing Systems <i>Data Structures & Algorithms, Operating Systems, Probability & Statistics, Operations Research</i>	Sep 2022 – Sep 2025

SELECTED PROJECTS

Adaptive Space-Time Finite Element Solver Dec 2025 – Jan 2026

- Engineered a distributed parallel solver for heat equation PDEs using the **deal.II** library and **MPI**.
- Implemented space-time adaptive mesh refinement to optimize computational load across distributed nodes, significantly reducing execution time for complex finite element simulations.
- Stack: C++, MPI, deal.II, HPC, Distributed Systems.*

Advanced Monte Carlo Integration & Optimization Dec 2025 – Jan 2026

- Built a high-performance, multithreaded library for N-dimensional Monte Carlo integration and MCMC.
- Leveraged **OpenMP** for parallelization, heavily optimizing cache locality to maximize **CPU core utilization**.
- Stack: C++, OpenMP, Stochastic Optimization, Cache Optimization.*

Client-Server Multiplayer Game Feb 2025 – Jun 2025

- Collaborated in a 4-person team to architect the complete **Model** and **Controller** layers (**MVC pattern**) for a real-time multiplayer digital board game.
- Managed state synchronization and multi-threading through **RMI** and **TCP/IP Sockets**, achieving 97% **JUnit** test coverage.
- Implemented a Graphical User Interface using **JavaFX** and **SceneBuilder**.
- Stack: Java, Sockets, MVC, Maven, JUnit, JavaFX.*

High-Performance CUDA Batched Matrix Multiplication Oct 2026

- Developed a highly optimized custom **CUDA** kernel to accelerate batched matrix multiplications.
- Maximized memory bandwidth and computational throughput by implementing **shared memory tiling**, **thread coarsening**, and **bank conflict avoidance** strategies.
- Stack: C/C++, CUDA, GPU Computing, Memory Profiling.*

Declarative Persistent NixOS System – Personal Daily Driver Apr 2026 – Present

- Architected and maintained a fully reproducible **Linux** environment using a **Btrfs** subvolume layout with snapshot-based root rollback (“erase your darlings” on-disk, not in-RAM), automatically restoring the root subvolume to a clean blank snapshot on every boot.
- Managed system packages and user dotfiles declaratively via **Nix Flakes** and **Home Manager**, with explicit persistence paths (via **impermanence**) selectively bind-mounting state that must survive reboots, enabling zero-touch provisioning and fully version-controlled system state.
- Stack: Nix, NixOS, Btrfs, Home Manager, Linux SysAdmin, Git.*

SKILLS

Languages	C, C++, Python, Java, Nix, Bash, SQL
HPC & Parallel	CUDA, MPI, OpenMP, GPU Computing, Cluster environments
Systems & Tools	Linux (Ubuntu, Arch, NixOS declarative config), Git, Docker, CMake, Slurm
ML & Math	PyTorch, Eigen, deal.II, Numerical Analysis, Pandas, Polars
Spoken	Italian (Native), English (C1 Equivalent – TOEIC: 985/990)